Git & GitHub.

Why Git & GitHub?

- •“When multiple developers work on the same project, managing code manually is very hard.”
- •“Git helps track changes, so we can always go back to older versions.”
- •“GitHub makes teamwork easy by storing code in the cloud.”
- •“Every software company uses Git/GitHub — so it’s a must-have skill.”

What is Git?

“Git is a Version Control System — think of it like Google Docs but for code.”

“Every time you make a change, Git can record it as a snapshot.”

“It works offline, everything is stored in your local computer.”

“Once you’re ready, you can sync changes with GitHub.”

Without Version Control

Oka writer *Pushpa 2* script rasthunnadu ani imagine chey.
Day 1 lo “Pushpa goes to forest” ani rasadu.
Day 10 lo mind marchi “Pushpa goes to city” ani modify chesadu.
Old version kavali ante… ayyo! malli dorakadu. Files overwrite ayyayi.
👉 Old versions save chesukolekapoyadam = **No Version Control**.

With Version Control (Git)

Ee same writer Git use chesthe:

Commit 1 → Pushpa goes to forest 🌲

Commit 2 → Pushpa goes to city 🌆

Commit 3 → Pushpa goes abroad ✈️
👉 ippudu writer ki anni versions record untayi.
Old story kavali ante commit history lo open chesukovachu.
Alane team lo inko writer ochi suggestions cheppina branches create chesi try cheyyachu.

Dev - 1

Dev - 2

Dev - 3

login page

Home page

Menus

Facebook

RPO

project repository

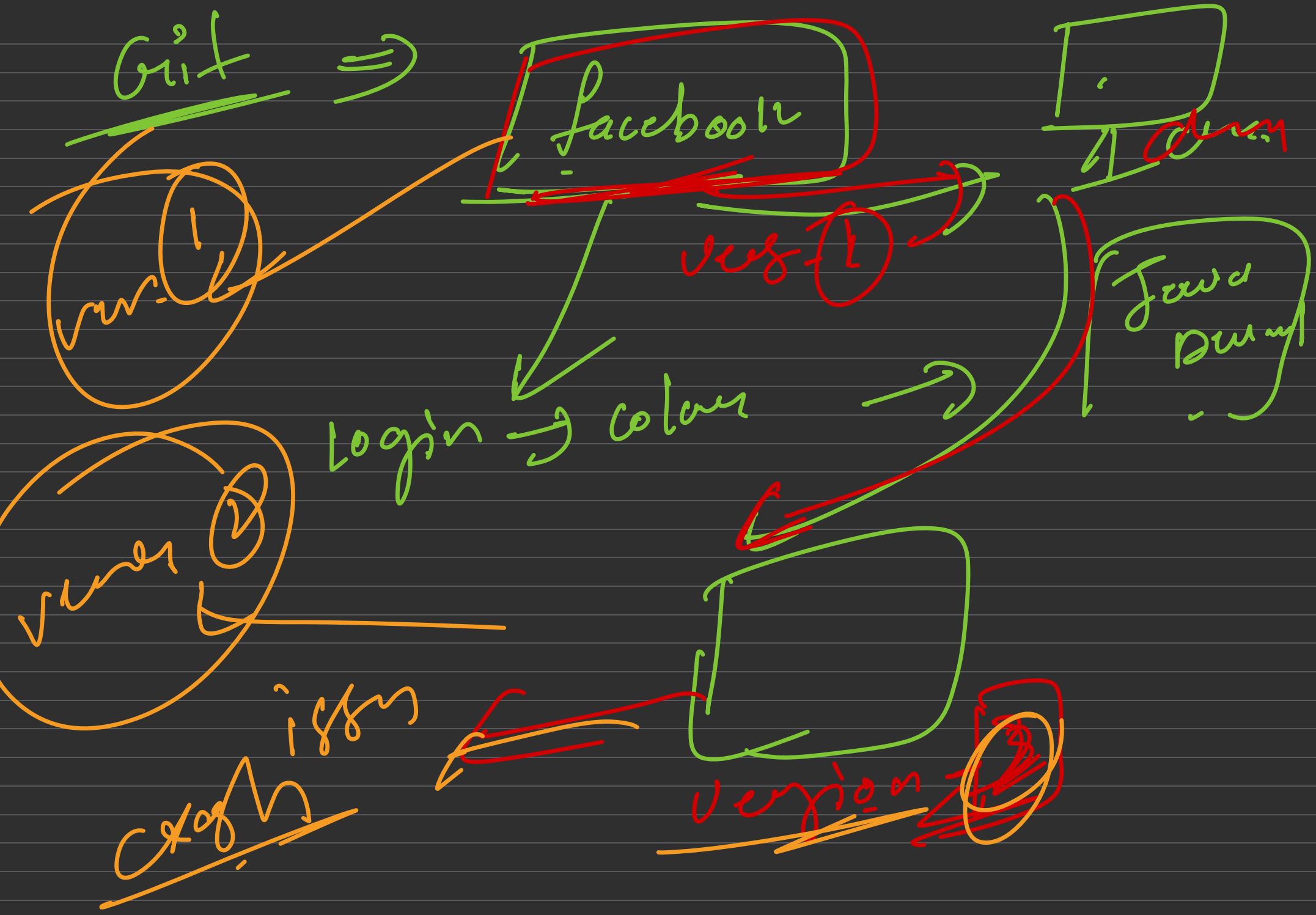
What is GitHub?

👉 *Notes*:

- •"GitHub is like social media for code."
- •"It stores your Git repositories in the cloud."
- •"You can share your projects, contribute to open-source, and showcase your portfolio."
- •"Features like Issues, Pull Requests, and Actions help in real teamwork."

Git hub → Repository → project

Git → local → stability


Git
facebook
ver 1
login → colour
ver 2
issues
version

Installing Git

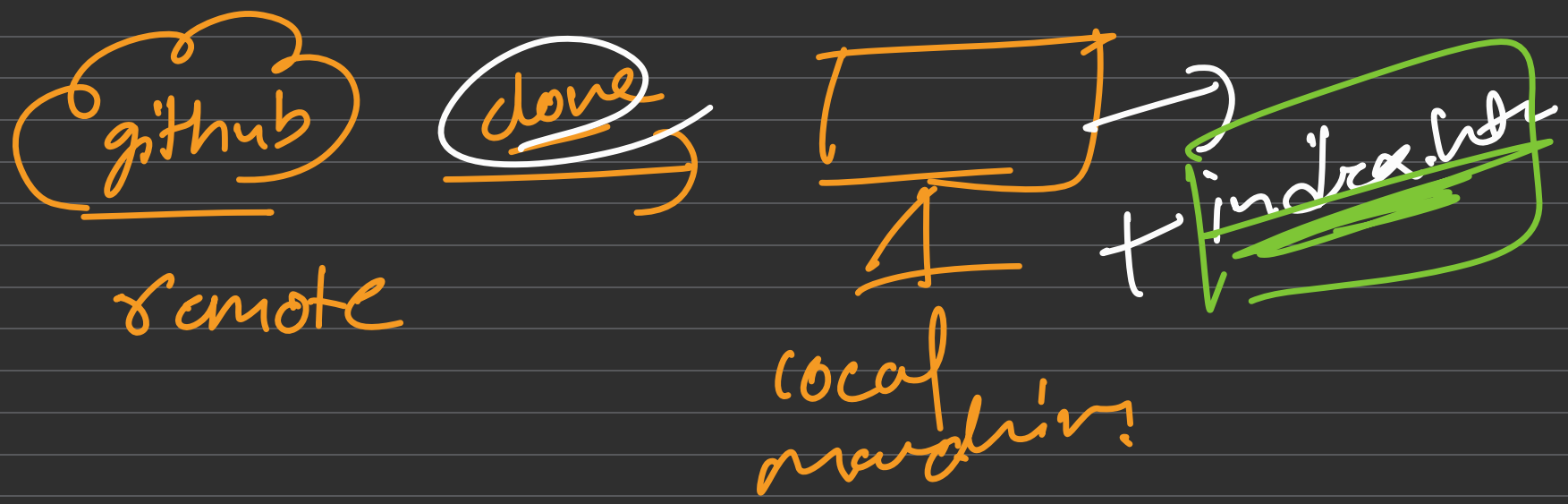
Commands

1.

Clone → copy the project from github to local system

git clone

Status → current status of the changes



status

git add filem

untracked

new files still not recognized

login.htm

git

modified

changed

file creation

existing file changes

staged

ready to be committed

unmodified

not changed

git & ~~trand~~

staged

add → stage → commit

git add

git commit

push

remote (github)

Git add

1. git add file_name
2. git add .
3. git add -v

Commit

git commit -m "msg"

init command

github new.

init — used to create a new git repo from terminal

1. git init
2. git remote add origin "<— link —>"
3. git remote -v
4. git branch
5. git branch -M main
6. git push origin main

repository new

changes

add

commit

login.htm

read

SimpayAP

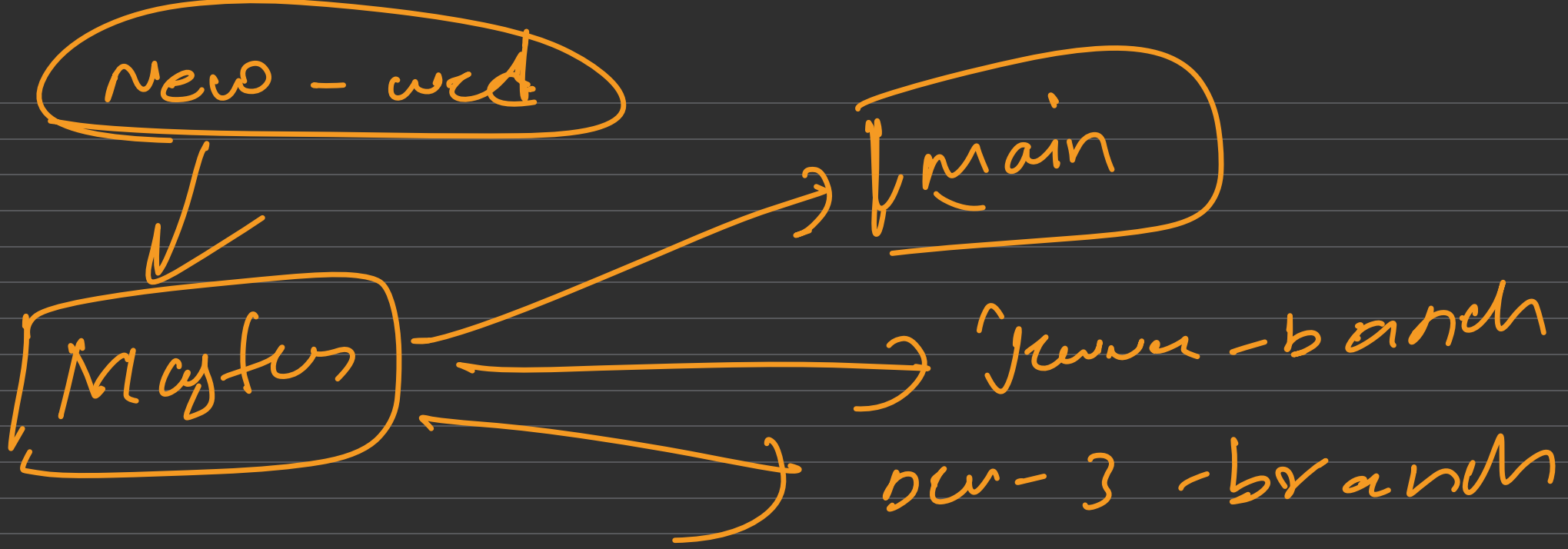
Remote

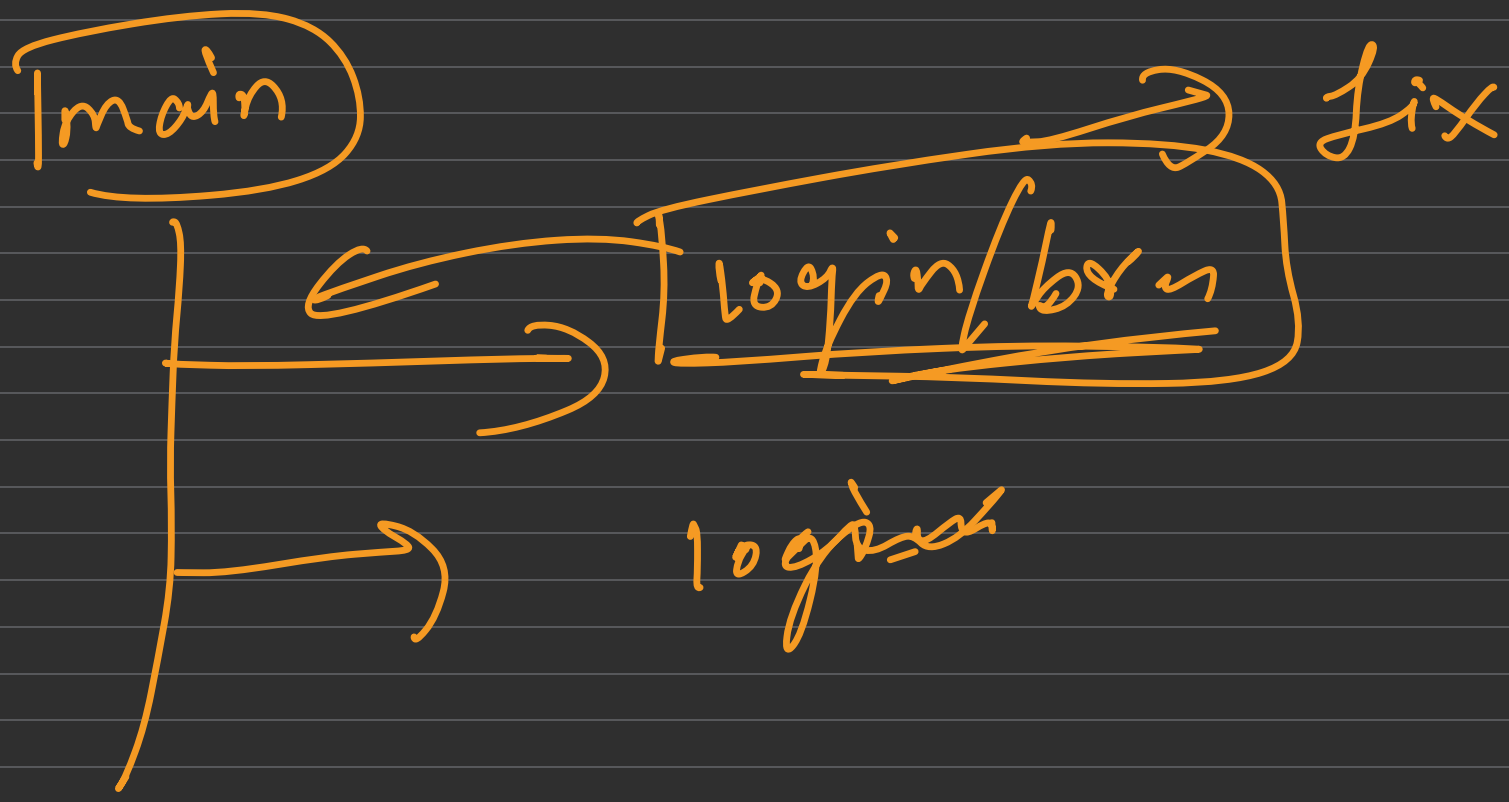
pw-2

Read index

Read index

Jira local pc


New - wcd
Master
main
Jenny-branch
dev-3-branch


main
fix
login/bra
login

WORK Flow

local Git

origin branch

repository

Git hub [remote]

clone

add

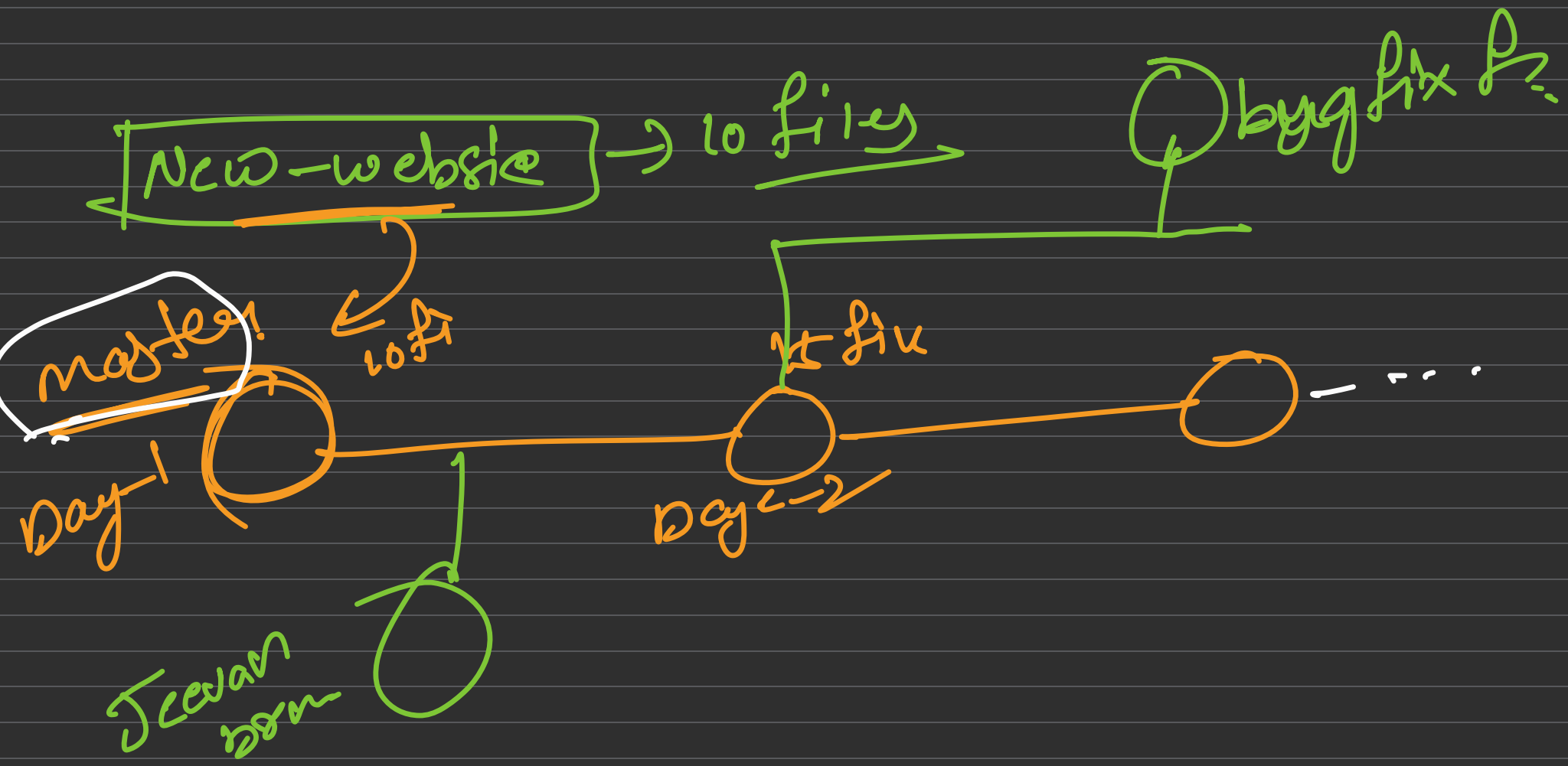
commit

push

branch

Master → default

our branch



branch commands

1. git branch
2. git checkout branch name
3. git branch -d <branch name>
4. git checkout -b <new branch name>
5. gif

Jeevanbasu

index.ht
Dreader
logout.ht

master

index.ht
greet

read-webpage

index
pen

branch 1

index.htm

Read

homepay.htm

branc 2

index.ht

rem

Ramu

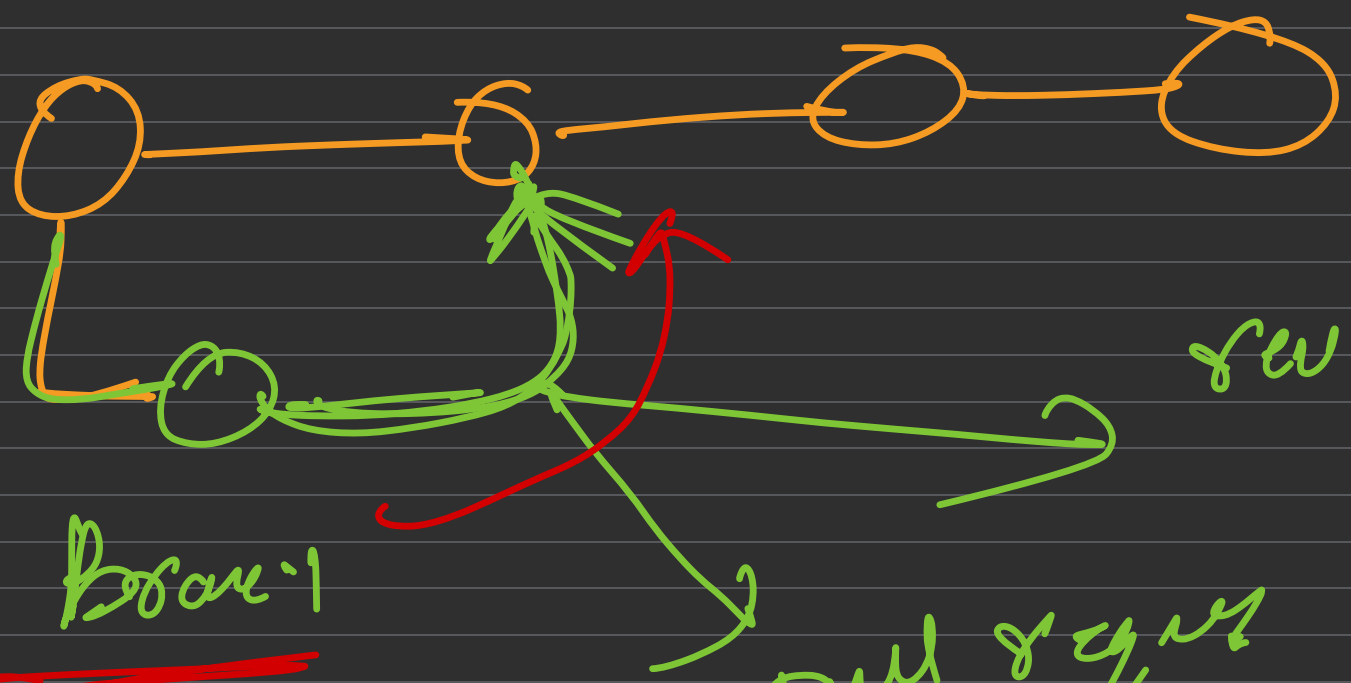
index

nus

Pull request

exist

master



review

Branch

pull request

new changes add

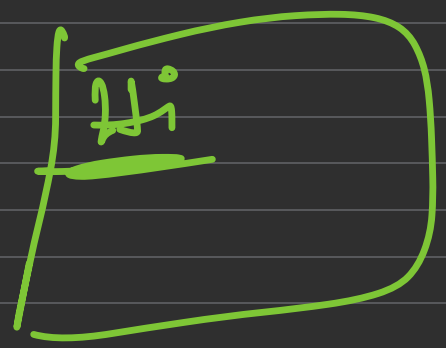
Merge conflicts

↳ occurs when there is similar change in same files from 2 different branches while merging.

branch 1

index.htm

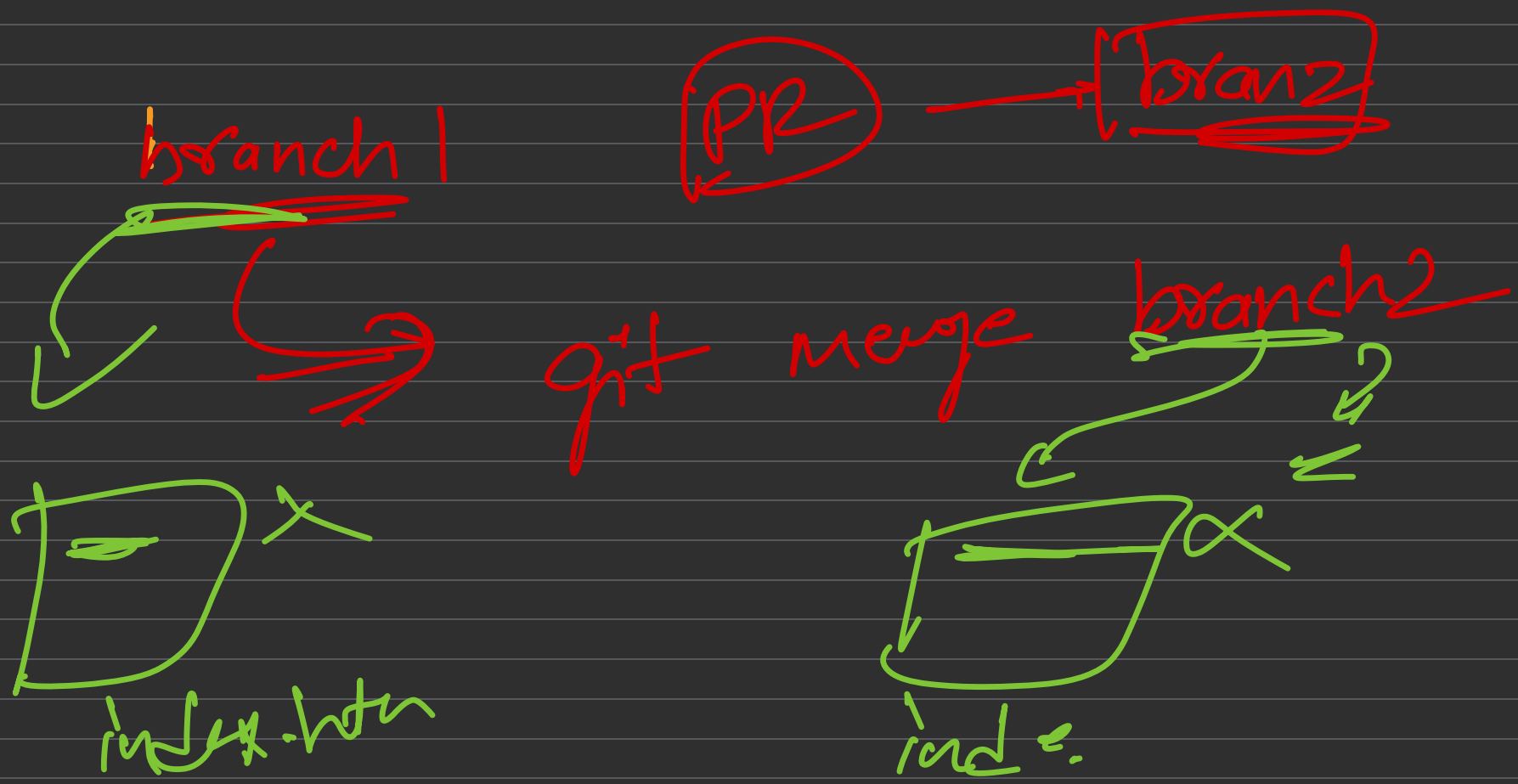
master



index.htm

merging

→ git merge branchname.



Undoing commands

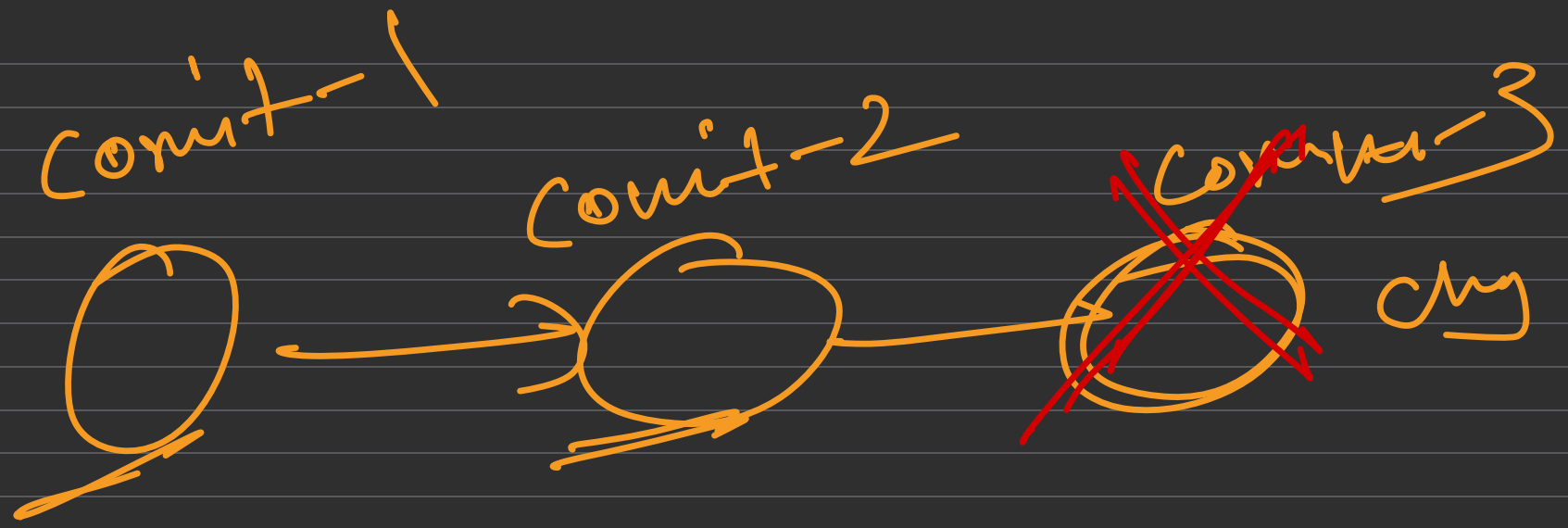
1. git reset <file name> → staged changes
   git reset
2. committed changes (for one commit)
   git reset HEAD~1
3. for multiple commits
   git reset <commit hash>
   git reset --hard <commit hash>

log.htm

unstaged →

git add

login.ht → stage → add → commit




```
git reset HEAD~1
```

Fork

→ a rough copy of the project

→ new repository that share code and visibility.

End note

1. flow →
2. local → repesit
3. repesit → local → push
4. commit Basic
5. branch → com
6. merging → conflic

remote
↓ cole
→ add
→ commit
→ push